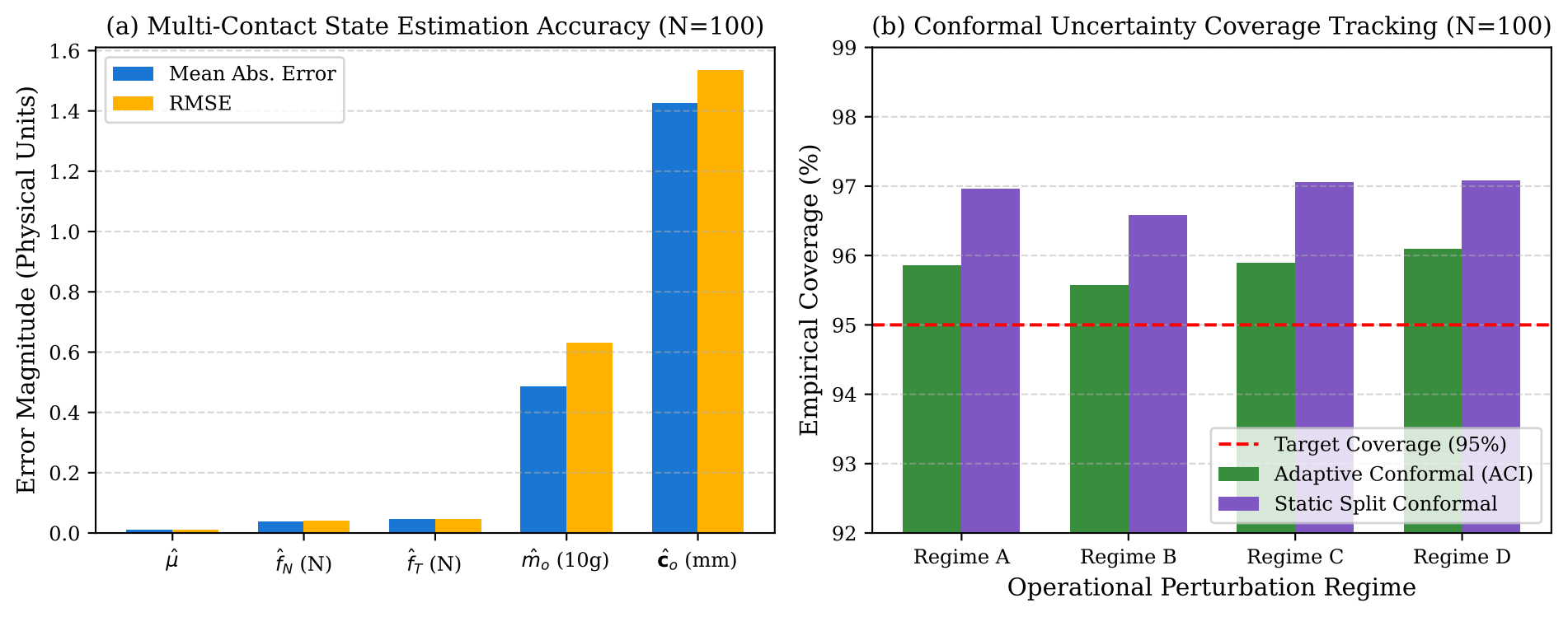
(a) Multi-Contact State Estimation Accuracy (N=100)
Error Magnitude (Physical Units)
Mean Abs. Error
RMSE
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
$\hat{\mu}$
$\hat{f}_N$ (N)
$\hat{f}_T$ (N)
$\hat{m}_o$ (10g)
$\hat{\mathbf{c}}_o$ (mm)
(b) Conformal Uncertainty Coverage Tracking (N=100)
Empirical Coverage (%)
92
93
94
95
96
97
98
99
Target Coverage (95%)
Adaptive Conformal (ACI)
Static Split Conformal
Regime A
Regime B
Regime C
Regime D
Operational Perturbation Regime